

StreamSense Database Schema

Viewer Engagement & Retention Insights

This document summarizes the main MongoDB collections used in the StreamSense application.

Users

Field	Type
user_id (PK)	ObjectId
name	String
email	String
password_hash	String
role	String

Content

Field	Type
content_id (PK)	ObjectId
title	String
genre	String
type	String

ViewerSession

Field	Type
session_id (PK)	ObjectId
user_id (FK)	ObjectId
content_id (FK)	ObjectId
watch_duration	Number
pause_count	Number
completion_%	Number

Analytics

Collection	Key Fields
EngagementScore	engagement_id, content_id, score
Retention	retention_id, user_id, renewed
Recommendation	recommendation_id, content_id, reason

Report	report_id, generated_by, report_type
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Relationships

- User (1) → ViewerSession (M)
- Content (1) → ViewerSession (M)
- Content (1) → EngagementScore (1)
- User (1) → Retention (1)
- Content (1) → Recommendation (M)
- Reports are generated from engagement, retention and recommendation data.